lecture_06.py

```python
import time
from typing import Callable
import torch
import torch.nn as nn
from torch.profiler import ProfilerActivity
from torch.utils.cpp_extension import load_inline
import triton
import triton.language as tl
from execute_util import text, link, image
from file_util import ensure_directory_exists
from lecture_util import article_link
from torch_util import get_device
from lecture_06_utils import check_equal, check_equal2, get_local_url, round1, mean
import os

def main():
    announcements()
```

Last lecture: high-level overview of GPUs and performance
This lecture: benchmarking/profiling + write kernels

```python
    if not torch.cuda.is_available():
        text("You should run this lecture on a GPU to get the full experience.")

    review_of_gpus()
    benchmarking_and_profiling()  # Important for understanding!

    kernel_fusion_motivation()
    cuda_kernels()  # Write kernels in CUDA/C++
    triton_kernels()  # Write kernels in Python
    pytorch_compilation()  # Don't write kernels at all?

    # More advanced computations
    triton_softmax_main()
```

Summary

Gap between the programming model (PyTorch, Triton, PTX) and hardware => performance mysteries

Benchmarking for understanding scaling
Profiling for understanding internals of PyTorch functions (bottoms out with kernels)
Looking at PTX assembly to understand internals of CUDA kernels

5 ways to write a function: manual, PyTorch, compiled, CUDA, Triton
GeLU (element-wise), softmax (row-wise), matmul (complex aggregation)

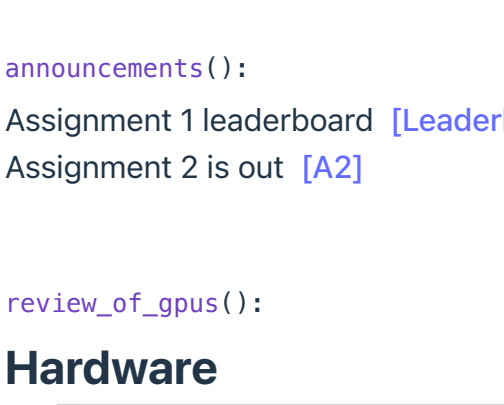
Key principle: organize computation to minimize reads/writes
Key ideas: kernel fusion (warehouse/factory analogy), tiling (shared memory)
Automatic compilers (Triton, torch.compile) will get better over time

```python
    further_reading()


def announcements():
```

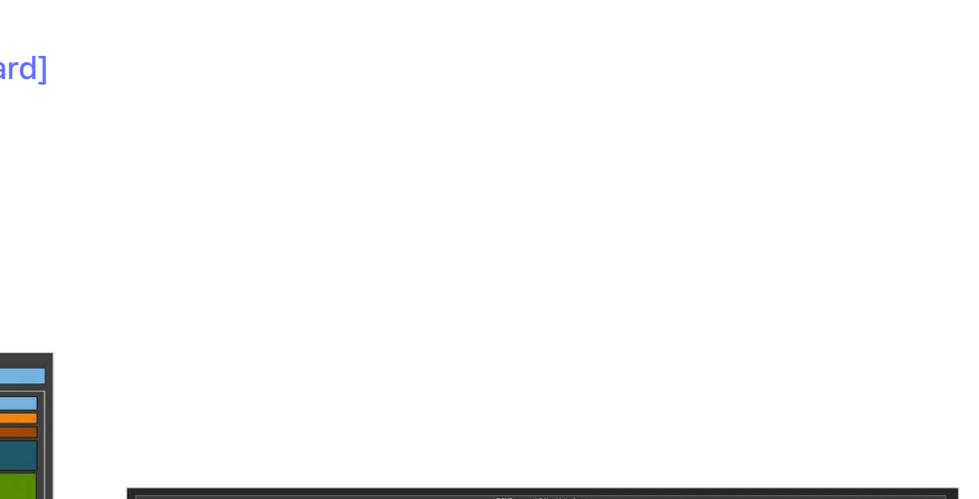
Assignment 1 leaderboard [Leaderboard]
Assignment 2 is out [A2]

```python
def review_of_gpus():
```

Hardware



SM



GA100 Full GPU with 128 SMs

Compute: streaming multiprocessors (SMs) [A100: 108]
Memory:

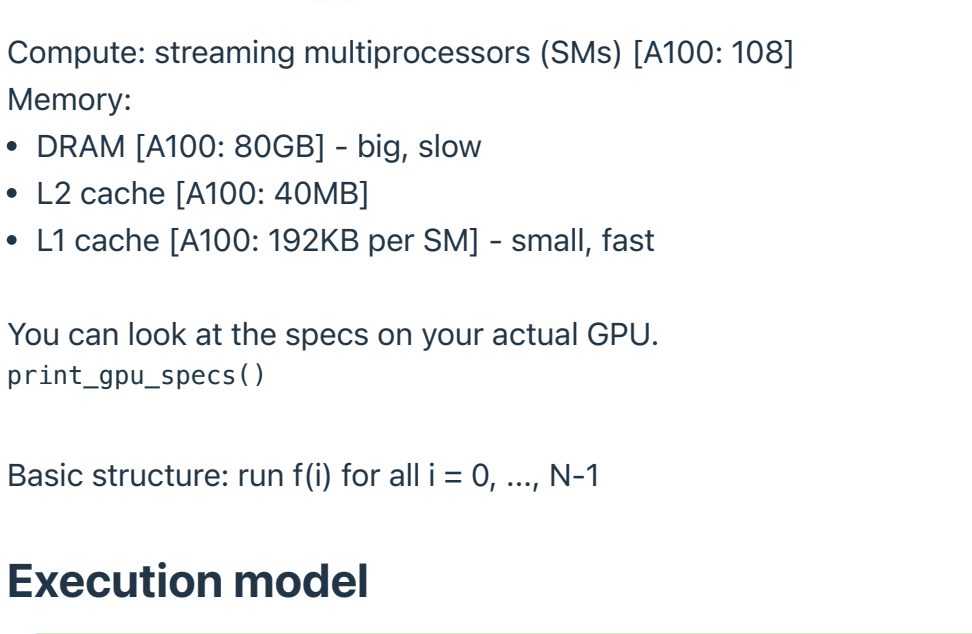
- DRAM [A100: 80GB] - big, slow
- L2 cache [A100: 40MB]
- L1 cache [A100: 192KB per SM] - small, fast

You can look at the specs on your actual GPU.

```python
    print_gpu_specs()
```

Basic structure: run f(i) for all i = 0, ..., N-1

Execution model

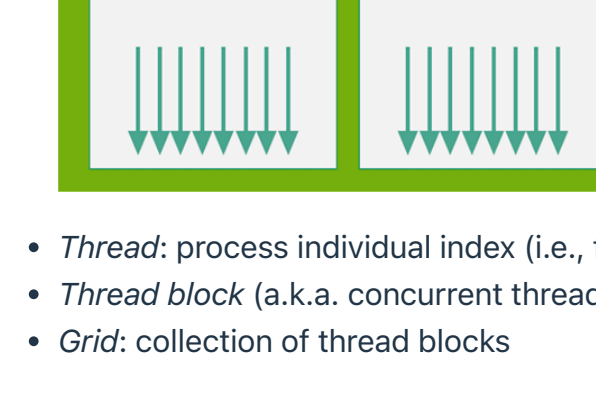


- *Thread*: process individual index (i.e., f(i))
- *Thread block* (a.k.a. concurrent thread arrays): scheduled on a single SM
- *Grid*: collection of thread blocks

Why thread blocks? Shared memory.

- Intuition: group f(i)'s that read similar data together
- Threads within a thread block have shared memory (as fast as L1 cache) [A100: 164KB]
- Can synchronize threads (for reading/writing) within a block (but not across blocks)

Hardware and execution interact.



Thread blocks scheduled onto SMs in waves.
Problem: last wave has fewer thread blocks, leaving some SMs idle (low occupancy).
Wave quantization: make number of thread blocks divide # SMs.
Rule of thumb: number of thread blocks should be >= 4x # SMs
Challenge: some aspects of hardware are hidden from the execution model (e.g., scheduling, # SMs).

Arithmetic intensity: # FLOPs / # bytes

- If high, operation is compute-bound (good)
- If low, operation is memory-bound (bad)

General rule: matrix multiplication is compute-bound, everything else is memory-bound

```python
def benchmarking_and_profiling():
```

IMPORTANT: benchmark/profile your code!

You can read spec sheets (marketing material) and papers
...but performance depends on your library version, your hardware, your workload
...so there is no substitute for benchmarking/profiling your code.

Example computation: running forward/backward passes on an MLP.

```python
    run_mlp(dim=128, num_layers=16, batch_size=128, num_steps=5)

    benchmarking()       # How long does it take?
    profiling()          # Where time is being spent?
```

Every time you make a change, benchmark/profile!

```python
class MLP(nn.Module):
    """Simple MLP: linear -> GeLU -> linear -> GeLU -> ... -> linear -> GeLU"""
    def __init__(self, dim: int, num_layers: int):
        super().__init__()
        self.layers = nn.ModuleList([nn.Linear(dim, dim) for _ in range(num_layers)])

    def forward(self, x: torch.Tensor):
        for layer in self.layers:
            x = layer(x)
            x = torch.nn.functional.gelu(x)
        return x


def run_mlp(dim: int, num_layers: int, batch_size: int, num_steps: int) -> Callable:
    # Define a model (with random weights)
    model = MLP(dim, num_layers).to(get_device())

    # Define an input (random)
    x = torch.randn(batch_size, dim, device=get_device())

    def run():
        # Run the model `num_steps` times (note: no optimizer updates)
        for step in range(num_steps):
            # Forward
            y = model(x).mean()

            # Backward
            y.backward()

    return run


def run_operation1(dim: int, operation: Callable) -> Callable:
    # Setup: create one random dim x dim matrices
    x = torch.randn(dim, dim, device=get_device())
    # Return a function to perform the operation
    return lambda : operation(x)


def run_operation2(dim: int, operation: Callable) -> Callable:
    # Setup: create two random dim x dim matrices
    x = torch.randn(dim, dim, device=get_device())
    y = torch.randn(dim, dim, device=get_device())
    # Return a function to perform the operation
    return lambda : operation(x, y)


def benchmarking():
```

Benchmarking measures the wall-clock time of performing some operation.

It only gives you end-to-end time, not where time is spent (profiling).

It is still useful for:

- comparing different implementations (which is faster?), and
- understanding how performance scales (e.g., with dimension).

Let's define a convenient function for benchmarking an arbitrary function.

```python
    benchmark("sleep", lambda : time.sleep(50 / 1000))
```

Benchmarking matrix multiplication

First, let us benchmark matrix multiplication of square matrices.

```python
    if torch.cuda.is_available():
        dims = (1024, 2048, 4096, 8192, 16384)  # @inspect dims
    else:
        dims = (1024, 2048)  # @inspect dims

    matmul_results = []
    for dim in dims:
        # @ inspect dim
        result = benchmark(f"matmul(dim={dim})", run_operation2(dim=dim, operation=lambda a, b: a @ b))
        matmul_results.append((dim, result))  # @inspect matmul_results
```

Let us benchmark our MLP!

```python
    dim = 256  # @inspect dim
    num_layers = 4  # @inspect num_layers
    batch_size = 256  # @inspect batch_size
    num_steps = 2  # @inspect num_steps

    mlp_base = benchmark("run_mlp", run_mlp(dim=dim, num_layers=num_layers, batch_size=batch_size, num_steps=num_steps)) # @inspect mlp_base
```

Scale the number of steps.

```python
    step_results = []
    for scale in (2, 3, 4, 5):
        result = benchmark(f"run_mlp({scale}x num_steps)",
                           run_mlp(dim=dim, num_layers=num_layers,
                                   batch_size=batch_size, num_steps=scale * num_steps)) # @inspect result, @inspect scale, @inspect num_steps
        step_results.append((scale, result))  # @inspect step_results
```

Scale the number of layers.

```python
    layer_results = []
    for scale in (2, 3, 4, 5):
        result = benchmark(f"run_mlp({scale}x num_layers)",
                           run_mlp(dim=dim, num_layers=scale * num_layers,
                                   batch_size=batch_size, num_steps=num_steps)) # @inspect result, @inspect scale, @inspect num_layers, @inspect num_steps
        layer_results.append((scale, result))  # @inspect layer_results
```

Scale the batch size.

```python
    batch_results = []
    for scale in (2, 3, 4, 5):
        result = benchmark(f"run_mlp({scale}x batch_size)",
                           run_mlp(dim=dim, num_layers=num_layers,
                                   batch_size=scale * batch_size, num_steps=num_steps)) # @inspect result, @inspect scale, @inspect num_layers, @inspect num_steps
        batch_results.append((scale, result))  # @inspect batch_results
```

Scale the dimension.

```python
    dim_results = []
    for scale in (2, 3, 4, 5):
        result = benchmark(f"run_mlp({scale}x dim)",
                           run_mlp(dim=scale * dim, num_layers=num_layers,
                                   batch_size=batch_size, num_steps=num_steps)) # @inspect result, @inspect scale, @inspect num_layers, @inspect num_steps
        dim_results.append((scale, result))  # @inspect dim_results
```

The timings are not always predictable due to the non-homogenous nature of CUDA kernels, hardware, etc.

You can also use `torch.utils.benchmark`, which provides more amenities.
https://pytorch.org/tutorials/recipes/recipes/benchmark.html
We did not use this to make benchmarking more transparent.

```python
def benchmark(description: str, run: Callable, num_warmups: int = 1, num_trials: int = 3):
    """Benchmark `func` by running it `num_trials`, and return all the times."""
    # Warmup: first times might be slower due to compilation, things not cached.
    # Since we will run the kernel multiple times, the timing that matters is steady state.
    for _ in range(num_warmups):
        run()
    if torch.cuda.is_available():
        torch.cuda.synchronize()  # Wait for CUDA threads to finish (important!)

    # Time it for real now!
    times: list[float] = [] # @inspect times, @inspect description
    for trial in range(num_trials):  # Do it multiple times to capture variance
        start_time = time.time()

        run()  # Actually perform computation
        if torch.cuda.is_available():
            torch.cuda.synchronize()  # Wait for CUDA threads to finish (important!)

        end_time = time.time()
        times.append((end_time - start_time) * 1000) # @inspect times

    mean_time = mean(times) # @inspect mean_time
    return mean_time


def profiling():
```

While benchmarking looks at end-to-end time, profiling looks at where time is spent.
Obvious: profiling helps you understand where time is being spent.
Deeper: profiling helps you understand (what is being called).

PyTorch has a nice built-in profiler https://pytorch.org/tutorials/recipes/recipes/profiler_recipe.html

Let's profile some code to see what is going on under the hood.

```python
    sleep_function = lambda : time.sleep(50 / 1000)
    sleep_profile = profile("sleep", sleep_function)
```

sleep

```
                 Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg    # of Calls

cudaDeviceSynchronize       100.00%      11.610us       100.00%      11.610us       5.805us             2

 Self CPU time total: 11.610us
```

Let's start with some basic operations.

```python
    add_function = lambda a, b: a + b
    add_profile = profile("add", run_operation2(dim=2048, operation=add_function))
```

add

```
                                                          Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se

                                                     aten::add        98.02%       1.392ms        99.38%       1.411ms       1.411ms      17.119us
 void at::native::vectorized_elementwise_kernel<4, at::native::CUDAFunctor_add...         0.00%       0.000us         0.00%       0.000us       0.000us      17.119us
                                              cudaLaunchKernel         1.37%      19.392us         1.37%      19.392us      19.392us       0.000us

                                         cudaDeviceSynchronize         0.62%       8.734us         0.62%       8.734us       4.367us       0.000us

 Self CPU time total: 1.420msSelf CUDA time total: 17.119us
```

```python
    matmul_function = lambda a, b: a @ b
    matmul_profile = profile("matmul", run_operation2(dim=2048, operation=matmul_function))
```

matmul

```
                                                          Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se

                                                  aten::matmul         2.29%       7.520us        97.24%     318.634us     318.634us       0.000us

                                                      aten::mm        90.14%     295.387us        94.94%     311.114us     311.114us     342.620us
 void cutlass::Kernel2(cutlass_80...         0.00%       0.000us         0.00%       0.000us       0.000us     342.620us       100.00%     342.620us     342.620us
                                       cudaDeviceGetAttribute         0.21%       0.690us         0.21%       0.690us       0.690us       0.000us

                                                cuLaunchKernel         4.59%      15.037us         4.59%      15.037us      15.037us       0.000us

                                         cudaDeviceSynchronize         2.76%       9.051us         2.76%       9.051us       4.525us       0.000us

 Self CPU time total: 327.685usSelf CUDA time total: 342.620us
```

```python
    matmul_function_128 = lambda a, b: a @ b
    matmul_profile_128 = profile("matmul(dim=128)", run_operation2(dim=128, operation=matmul_function_128))
```

matmul(dim=128)

```
                                                          Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se

                                                  aten::matmul         1.17%       4.912us        98.24%     413.723us     413.723us       0.000us

                                                      aten::mm        42.40%     178.581us        97.07%     408.811us     408.811us       4.992us
 sm80_xmma_gemm_f32f32_f32f32_f32_nn_n_tilesize32x32x8_stage3_warpsize1x2x1_ff...         0.00%       0.000us         0.00%       0.000us       0.000us       4.992us
                                        cudaFuncGetAttributes         0.96%       4.023us         0.96%       4.023us       4.023us       0.000us

                                           cudaLaunchKernelExC        53.71%     226.207us        53.71%     226.207us     226.207us       0.000us

                                         cudaDeviceSynchronize         1.76%       7.413us         1.76%       7.413us       3.707us       0.000us

 Self CPU time total: 421.136usSelf CUDA time total: 4.992us
```

Observations

- You can see what CUDA kernels are actually being called.
- Different CUDA kernels are invoked depending on the tensor dimensions.

Name of CUDA kernel tells us something about the implementation.
Example: cutlass_80_simt_sgemm_256x128_8x4_nn_align1

- cutlass: NVIDIA's CUDA library for linear algebra
- 256x128: tile size

Let's now look at some composite operations.

```python
    cdist_function = lambda a, b: torch.cdist(a, b)
    cdist_profile = profile("cdist", run_operation2(dim=2048, operation=cdist_function))
```

cdist

```
                                                          Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se

                                                   aten::cdist         1.38%      27.430us        99.62%       1.982ms       1.982ms       0.000us

                                          aten::_euclidean_dist         2.92%      58.128us        97.28%       1.936ms       1.936ms       0.000us

                                                  aten::matmul         0.10%       1.961us         2.51%      49.898us      49.898us       0.000us

                                                      aten::mm         1.92%      38.220us         2.41%      47.937us      47.937us     343.740us
 sm80_xmma_gemm_f32f32_f32f32_f32_tn_n_tilesize128x128x8_stage3_warpsize2x2x1_...         0.00%       0.000us         0.00%       0.000us       0.000us     343.740us
                                                     aten::cat         0.88%      17.459us         1.33%      26.451us      13.226us      29.888us
 void at::native::(anonymous namespace)::CatArrayBatchedCopy_aligned16_contig<...         0.00%       0.000us         0.00%       0.000us       0.000us      29.888us
                                                     aten::pow        72.92%       1.451ms        84.00%       1.671ms     835.719us      22.304us
 void at::native::vectorized_elementwise_kernel<4, at::native::(anonymous name...         0.00%       0.000us         0.00%       0.000us       0.000us      22.304us
                                                     aten::sum         1.22%      24.211us         1.77%      35.272us      17.636us      14.976us

 Self CPU time total: 1.990msSelf CUDA time total: 440.121us
```

```python
    gelu_function = lambda a, b: torch.nn.functional.gelu(a + b)
    gelu_profile = profile("gelu", run_operation2(dim=2048, operation=gelu_function))
```

gelu

```
                                                          Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se

                                                     aten::add        86.27%       1.422ms        98.31%       1.620ms       1.620ms      18.304us
 void at::native::vectorized_elementwise_kernel<4, at::native::CUDAFunctor_add...         0.00%       0.000us         0.00%       0.000us       0.000us      18.304us
                                                    aten::gelu         0.74%      12.250us         1.13%      18.672us      18.672us       9.056us
 void at::native::vectorized_elementwise_kernel<4, at::native::GeluCUDAKernelI...         0.00%       0.000us         0.00%       0.000us       0.000us       9.056us
                                              cudaLaunchKernel        12.42%     204.811us        12.42%     204.811us     102.406us       0.000us

                                         cudaDeviceSynchronize         0.56%       9.236us         0.56%       9.236us       4.618us       0.000us

 Self CPU time total: 1.648msSelf CUDA time total: 27.360us
```

```python
    softmax_function = lambda a, b: torch.nn.functional.softmax(a + b, dim=-1)
    softmax_profile = profile("softmax", run_operation2(dim=2048, operation=softmax_function))
```

softmax

```
                                                          Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se

                                                 aten::softmax         0.70%      11.487us         1.85%      30.256us      30.256us       0.000us

                                                aten::_softmax         0.73%      11.951us         1.15%      18.769us      18.769us      20.191us
 void at::native::(anonymous namespace)::cunn_SoftMaxForwardSmem<4, float, flo...         0.00%       0.000us         0.00%       0.000us       0.000us      20.191us
                                                     aten::add        87.82%       1.434ms        97.71%       1.596ms       1.596ms      18.528us
 void at::native::vectorized_elementwise_kernel<4, at::native::CUDAFunctor_add...         0.00%       0.000us         0.00%       0.000us       0.000us      18.528us
                                              cudaLaunchKernel        10.31%     168.454us        10.31%     168.454us      84.227us       0.000us

                                         cudaDeviceSynchronize         0.43%       7.097us         0.43%       7.097us       3.549us       0.000us

 Self CPU time total: 1.633msSelf CUDA time total: 38.719us
```

Now let's profile our MLP.
We will also visualize our stack trace using a flame graph, which reveals where time is being spent.

```python
    if torch.cuda.is_available():
        mlp_profile = profile("mlp", run_mlp(dim=2048, num_layers=64, batch_size=1024, num_steps=2), with_stack=True)
    else:
        mlp_profile = profile("mlp", run_mlp(dim=128, num_layers=16, batch_size=128, num_steps=2), with_stack=True)
```

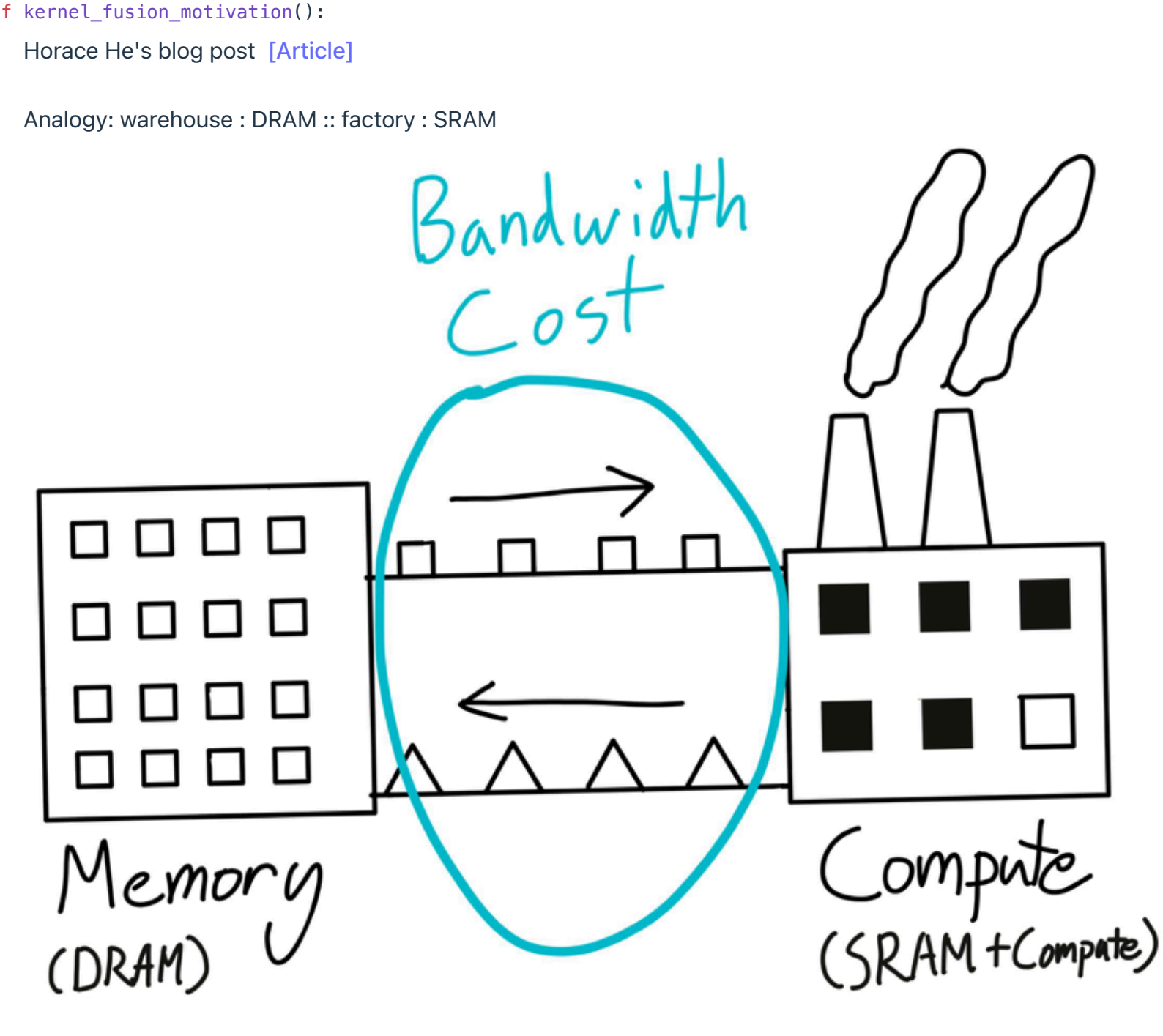
mlp

```
                                                          Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se

                      autograd::engine::evaluate_function: AddmmBackward0         1.72%     672.089us        16.79%       6.571ms      51.339us       0.000us

                                                AddmmBackward0         1.32%     517.317us        10.87%       4.253ms      33.224us       0.000us

                                                      aten::mm         5.47%       2.141ms         7.96%       3.115ms      12.263us      44.625ms

                                                  aten::linear         0.55%     217.124us        12.51%       4.895ms      38.245us       0.000us

                                                   aten::addmm         7.09%       2.773ms        10.67%       4.176ms      32.621us      23.450ms
 sm80_xmma_gemm_f32f32_f32f32_f32_tn_n_tilesize128x128x8_stage3_warpsize2x2x1_...         0.00%       0.000us         0.00%       0.000us       0.000us      23.450ms
        autograd::engine::evaluate_function: torch::autograd::AccumulateGrad         1.34%     522.511us         5.72%       2.238ms       8.743us       0.000us

                                   torch::autograd::AccumulateGrad         0.88%     342.816us         4.38%       1.716ms       6.702us       0.000us

 Self CPU time total: 39.129msSelf CUDA time total: 73.598ms
```

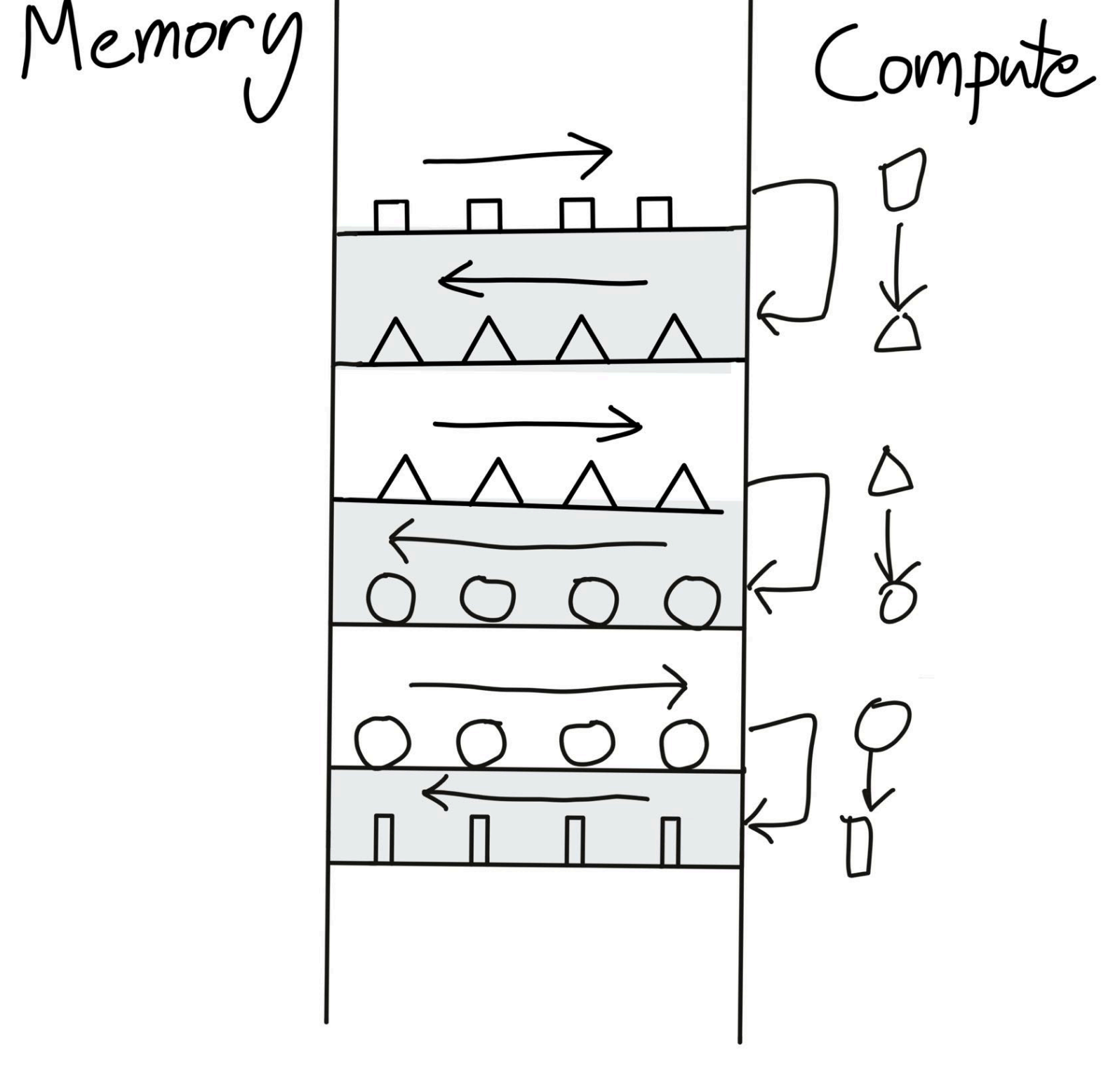
```python
def profile(description: str, run: Callable, num_warmups: int = 1, with_stack: bool = False):
    # Warmup
    for _ in range(num_warmups):
        run()
    if torch.cuda.is_available():
        torch.cuda.synchronize()  # Wait for CUDA threads to finish (important!)

    # Run the code with the profiler
    with torch.profiler.profile(
            activities=[ProfilerActivity.CPU, ProfilerActivity.CUDA],
            # Output stack trace for visualization
            with_stack=with_stack,
            # Needed to export stack trace for visualization
            experimental_config=torch._C._profiler._ExperimentalConfig(verbose=True)) as prof:
        run()
        if torch.cuda.is_available():
            torch.cuda.synchronize()  # Wait for CUDA threads to finish (important!)

    # Print out table
    table = prof.key_averages().table(sort_by="cuda_time_total",
                                      max_name_column_width=80,
                                      row_limit=10)
    #text(f"## {description}")
    #text(table, verbatim=True)

    # Write stack trace visualization
    if with_stack:
        text_path = f"var/stacks_{description}.txt"
        svg_path = f"var/stacks_{description}.svg"
        prof.export_stacks(text_path, "self_cuda_time_total")

    return table

def kernel_fusion_motivation():
```

Horace He's blog post [Article]

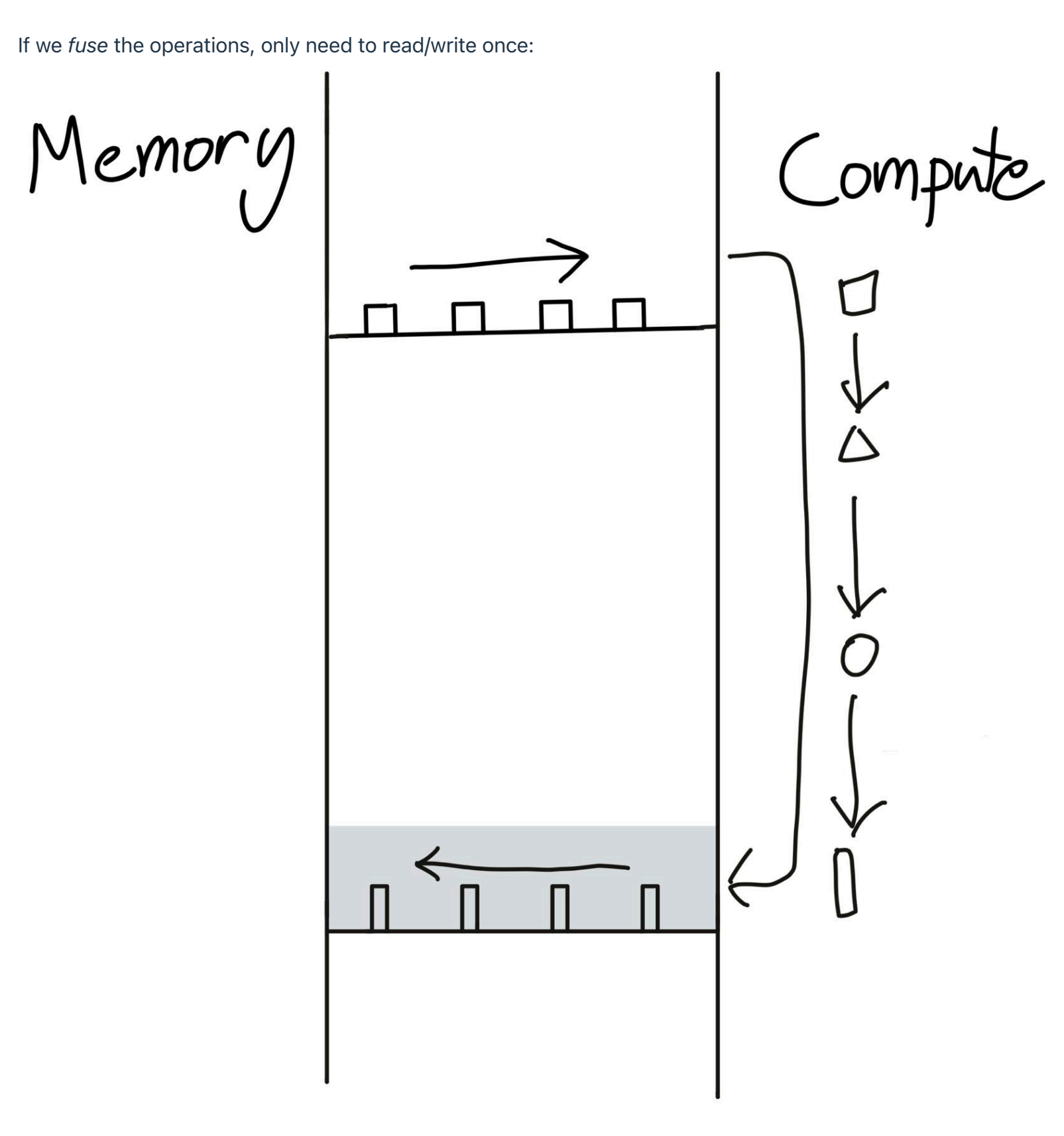
Analogy: warehouse : DRAM :: factory : SRAM



Each operation needs to read/compute/write:



If we *fuse* the operations, only need to read/write once:



To see the effect of fusion, let's consider the GeLU activation function.
https://pytorch.org/docs/stable/generated/torch.nn.GELU.html

Let's consider two ways to compute GeLU:

```
x = torch.tensor([1.])  # @inspect x
```

1. The default PyTorch implementation (fused):

```
y1 = pytorch_gelu(x)  # @inspect y1
```

2. We can also write our own by hand (not fused):

```
y2 = manual_gelu(x)  # @inspect y2

# Check that the implementations match
assert torch.allclose(y1, y2)

# Check more systematically
check_equal(pytorch_gelu, manual_gelu)
```

Let's benchmark.

```
manual_time = benchmark("manual_gelu", run_operation1(dim=16384, operation=manual_gelu)) # @inspect manual_time
pytorch_time = benchmark("pytorch_gelu", run_operation1(dim=16384, operation=pytorch_gelu)) # @inspect pytorch_time
if manual_time is not None and pytorch_time is not None:
```

The fused version is significantly faster: 8.15 ms, 1.13 ms

```
else:
    text("Could not compare times - benchmark results were None")
```

Let's look under the hood.

```
manual_gelu_profile = profile("manual_gelu", run_operation1(dim=16384, operation=manual_gelu))
```

manual_gelu

```
                                                                          Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se

                                                                     aten::mul        15.19%       1.479ms        26.01%       2.532ms     422.074us       5.222ms
 void at::native::vectorized_elementwise_kernel<4, at::native::BinaryFunctor<f...         0.00%       0.000us         0.00%       0.000us       0.000us       3.112ms
                                                                     aten::add         0.13%      12.473us         0.21%      20.089us      10.045us       1.742ms
 void at::native::vectorized_elementwise_kernel<4, at::native::CUDAFunctor_add...         0.00%       0.000us         0.00%       0.000us       0.000us       1.036ms
                                                                    aten::tanh         0.07%       6.961us         0.12%      11.588us      11.588us     704.343us
 void at::native::vectorized_elementwise_kernel<4, at::native::tanh_kernel_cud...         0.00%       0.000us         0.00%       0.000us       0.000us     704.343us
                                                              cudaLaunchKernel        10.95%       1.066ms        10.95%       1.066ms     118.432us       0.000us

                                                         cudaDeviceSynchronize        73.66%       7.171ms        73.66%       7.171ms       3.586ms       0.000us

 Self CPU time total: 9.735msSelf CUDA time total: 7.669ms
```

```
pytorch_gelu_profile = profile("pytorch_gelu", run_operation1(dim=16384, operation=pytorch_gelu))
```

pytorch_gelu

```
                                                                          Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se

                                                                    aten::gelu        71.12%       1.436ms        84.28%       1.701ms       1.701ms     701.560us
 void at::native::vectorized_elementwise_kernel<4, at::native::GeluCUDAKernelI...         0.00%       0.000us         0.00%       0.000us       0.000us     701.560us
                                                              cudaLaunchKernel        13.16%     265.687us        13.16%     265.687us     265.687us       0.000us

                                                         cudaDeviceSynchronize        15.72%     317.405us        15.72%     317.405us     158.703us       0.000us

 Self CPU time total: 2.019msSelf CUDA time total: 701.560us
```

The PyTorch just calls one kernel whereas the others are atomic (remember the warehouse/factory)

Look at Nsight profiler for MLP

```
def cuda_kernels():
```

Now let's open the box to understand what's going on inside a CUDA kernel by writing our own.

Let's write the GeLU function in CUDA.

```
cuda_gelu = create_cuda_gelu() # @inspect cuda_gelu
x = manual_gelu # @inspect x
```

Check correctness of our implementation.

```
if cuda_gelu is not None:
    check_equal(cuda_gelu, manual_gelu)
```

Benchmark our CUDA version.

```
pytorch_time = benchmark("pytorch_gelu", run_operation1(dim=16384, operation=pytorch_gelu)) # @inspect pytorch_time
manual_time = benchmark("manual_gelu", run_operation1(dim=16384, operation=manual_gelu)) # @inspect manual_time
if cuda_gelu is not None:
    cuda_time = benchmark("cuda_gelu", run_operation1(dim=16384, operation=cuda_gelu)) # @inspect cuda_time
    cuda_gelu_profile = profile("cuda_gelu", run_operation1(dim=16384, operation=cuda_gelu))
```

cuda_gelu

```
                           Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Self CUDA %    CUDA total  CUDA time avg

 gelu_kernel(float*, float*, int)         0.00%       0.000us         0.00%       0.000us       0.000us       1.664ms       100.00%       1.664ms       1.664ms

                aten::empty_like         0.20%       6.209us        46.52%       1.428ms       1.428ms       0.000us         0.00%       0.000us       0.000us

             aten::empty_strided        46.31%       1.422ms        46.31%       1.422ms       1.422ms       0.000us         0.00%       0.000us       0.000us

                cudaLaunchKernel         8.93%     274.078us         8.93%     274.078us     274.078us       0.000us         0.00%       0.000us       0.000us

           cudaDeviceSynchronize        44.56%       1.368ms        44.56%       1.368ms     683.944us       0.000us         0.00%       0.000us       0.000us

 Self CPU time total: 3.070msSelf CUDA time total: 1.664ms
```

Our CUDA implementation is faster than manual, but not as good as PyTorch.

Elementwise operations are easy in CUDA (though you can still be smarter).
But most interesting operations (e.g., matmul, softmax, RMSNorm) require reading multiple values.
For that, you have to think about managing shared memory, etc.

```
def create_cuda_gelu():
```

CUDA is an extension of C/C++ with APIs for managing GPUs.

Simplified picture: write f(i), CUDA kernel computes f(i) for all i.

Grid: collection of thread blocks: numBlocks = (2, 4), blockDim = (1, 8)
Thread block: collection of threads: blockIdx = (0, 1)
Thread: single unit of operation: threadIdx = (0, 3).

You write code that a thread execute, using (blockIdx, blockDim, threadIdx) to determine what to do.

Set CUDA_LAUNCH_BLOCKING so that if there are errors, CUDA will tell you what went wrong.

```
os.environ["CUDA_LAUNCH_BLOCKING"] = "1"
```

The `load_inline` function makes it convenient to write CUDA code and bind it to a Python module for immediate use.

```
# CUDA code: has the full logic
cuda_gelu_src = open("gelu.cu").read()
#include <math.h>#include <torch/extension.h>#include <c10/cuda/CUDAException.h>global void gelu_kernel(float* in, float* out, int num_elements) {

// Get the index into the tensor

int i = blockIdx.x * blockDim.x + threadIdx.x;

if (i < num_elements) {  // To handle the case when n < numBlocks * blockDim

    // Do the actual computation

    out[i] = 0.5 * in[i] * (1.0 + tanh(0.79788456 * (in[i] + 0.044715 * in[i] * in[i] * in[i])));

}

}inline unsigned int cdiv(unsigned int a, unsigned int b) {

// Compute ceil(a / b)

return (a + b - 1) / b;

}torch::Tensor gelu(torch::Tensor x) {

TORCH_CHECK(x.device().is_cuda());

TORCH_CHECK(x.is_contiguous());

// Allocate empty tensor

torch::Tensor y = torch::empty_like(x);

// Determine grid (elements divided into blocks)

int num_elements = x.numel();

int block_size = 1024;  // Number of threads

int num_blocks = cdiv(num_elements, block_size);

// Launch the kernel

gelu_kernel<<<num_blocks, block_size>>>(x.data_ptr<float>(), y.data_ptr<float>(), num_elements);

C10_CUDA_KERNEL_LAUNCH_CHECK();  // Catch errors immediately

return y;

}

# C++ code: defines the gelu function
cpp_gelu_src = "torch::Tensor gelu(torch::Tensor x);"
```

Compile the CUDA code and bind it to a Python module.

```
ensure_directory_exists("var/cuda_gelu")
if not torch.cuda.is_available():
    return None
module = load_inline(
    cuda_sources=[cuda_gelu_src],
    cpp_sources=[cpp_gelu_src],
    functions=["gelu"],
    extra_cflags=["-O2"],
    verbose=True,
    name="inline_gelu",
    build_directory="var/cuda_gelu",
)

cuda_gelu = getattr(module, "gelu")
return cuda_gelu


def triton_kernels():
    triton_introduction()
    triton_gelu_main()


def triton_introduction():
```

Developed by OpenAI in 2021
https://openai.com/research/triton

Make GPU programming more accessible
- Write in Python
- Think about thread blocks rather than threads

```
What does Triton offer?
                                          CUDA      Triton

• Memory coalescing (transfer from DRAM)     manual    automatic

• Shared memory management                   manual    automatic

• Scheduling within SMs                      manual    automatic

• Scheduling across SMs                      manual    manual
```

Compiler does more work, can actually outperform PyTorch implementations!

```
def triton_gelu_main():
    if not torch.cuda.is_available():
        return
```

One big advantage of Triton is that you can step through the Python code.

Let's step through a Triton kernel.

```
x = torch.randn(8192, device=get_device())
y1 = triton_gelu(x)

print_ptx_main()  # Look at the generated instructions
```

Check that it's correct.

```
check_equal(triton_gelu, manual_gelu)
```

Let's now benchmark it compared to the PyTorch and CUDA implementations.
Remember to set TRITON_INTERPRET=0 for good performance.

```
manual_time = benchmark("manual_gelu", run_operation1(dim=16384, operation=manual_gelu)) # @inspect manual_time
pytorch_time = benchmark("pytorch_gelu", run_operation1(dim=16384, operation=pytorch_gelu)) # @inspect pytorch_time
cuda_time = benchmark("cuda_gelu", run_operation1(dim=16384, operation=create_cuda_gelu())) # @inspect cuda_time
triton_time = benchmark("triton_gelu", run_operation1(dim=16384, operation=triton_gelu)) # @inspect triton_time

triton_gelu_profile = profile("triton_gelu", run_operation1(dim=16384, operation=triton_gelu))
```

triton_gelu

```
                   Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Self CUDA %    CUDA total  CUDA time avg    # of Calls

     triton_gelu_kernel         0.00%       0.000us         0.00%       0.000us       0.000us     705.240us       100.00%     705.240us     705.240us             1

       aten::empty_like         0.32%       5.629us        72.49%       1.267ms       1.267ms       0.000us         0.00%       0.000us       0.000us             1

    aten::empty_strided        72.16%       1.261ms        72.16%       1.261ms       1.261ms       0.000us         0.00%       0.000us       0.000us             1

         cuLaunchKernel        16.46%     287.632us        16.46%     287.632us     287.632us       0.000us         0.00%       0.000us       0.000us             1

  cudaDeviceSynchronize        11.05%     193.144us        11.05%     193.144us      96.572us       0.000us         0.00%       0.000us       0.000us             2

 Self CPU time total: 1.747msSelf CUDA time total: 705.240us
```

Our Triton implementation (triton_gelu):
- is almost as good as the PyTorch implementation (pytorch_gelu).
- is actually slower than our naive CUDA implementation (cuda_gelu).

Triton operates on blocks, CUDA operates on threads.
Blocks allows Triton compiler to do other optimizations (e.g., thread coarsening).

Everything is way faster than the manual implementation (manual_gelu).

```
def triton_gelu(x: torch.Tensor):
    assert x.is_cuda
    assert x.is_contiguous()

    # Allocate output tensor
    y = torch.empty_like(x)

    # Determine grid (elements divided into blocks)
    num_elements = x.numel()
    block_size = 1024  # Number of threads
    num_blocks = triton.cdiv(num_elements, block_size)

    triton_gelu_kernel[(num_blocks,)](x, y, num_elements, BLOCK_SIZE=block_size)

    return y


@triton.jit
def triton_gelu_kernel(x_ptr, y_ptr, num_elements, BLOCK_SIZE: tl.constexpr):
    # Input is at `x_ptr` and output is at `y_ptr`
    #     |        Block 0            |          Block 1          |      ...      |
    #                            BLOCK_SIZE                                 num_elements

    pid = tl.program_id(axis=0)
    block_start = pid * BLOCK_SIZE

    # Indices where this thread block should operate
    offsets = block_start + tl.arange(0, BLOCK_SIZE)

    # Handle boundary
    mask = offsets < num_elements

    # Read
    x = tl.load(x_ptr + offsets, mask=mask)

    # Approx gelu is 0.5 * x * (1 + tanh(sqrt(2/pi) * (x + 0.044715 * x^3)))
    # Compute (tl.tanh doesn't exist, use tanh(a) = (exp(2a) - 1) / (exp(2a) + 1)
    a = 0.79788456 * (x + 0.044715 * x * x * x)
    exp = tl.exp(2 * a)
    tanh = (exp - 1) / (exp + 1)
    y = 0.5 * x * (1 + tanh)

    # Store
    tl.store(y_ptr + offsets, y, mask=mask)


def print_ptx_main():
```

PTX (parallel thread execution) is like an assembly language for GPUs.

We can see the PTX code generated by Triton.
https://docs.nvidia.com/cuda/parallel-thread-execution/index.html

```
ptx = print_ptx("triton_gelu", triton_gelu_kernel)
//// Generated by LLVM NVPTX Back-End//.version 8.4.target sm_90a.address_size 64

// .globl       triton_gelu_kernel      // -- Begin function triton_gelu_kernel
                                        // @triton_gelu_kernel
.visible .entry triton_gelu_kernel(

.param .u64 .ptr .global .align 1 triton_gelu_kernel_param_0,

.param .u64 .ptr .global .align 1 triton_gelu_kernel_param_1,

.param .u32 triton_gelu_kernel_param_2

).reqntid 128, 1, 1{

.reg .pred      %p<5>;

.reg .b32       %r<49>;

.reg .f32       %f<113>;

.reg .b64       %rd<8>;

.loc    1 552 0                         // lecture_06.py:552:0

$L__func_begin0:

.loc    1 552 0                         // lecture_06.py:552:0

// %bb.0:

ld.param.u64    %rd5, [triton_gelu_kernel_param_0];

ld.param.u64    %rd6, [triton_gelu_kernel_param_1];

$L__tmp0:

.loc    1 557 24                        // lecture_06.py:557:24

// begin inline asm

mov.u32 %r1, %ctaid.x;

// end inline asm

.loc    1 558 24                        // lecture_06.py:558:24

shl.b32         %r42, %r1, 10;

ld.param.u32    %r43, [triton_gelu_kernel_param_2];

.loc    1 561 41                        // lecture_06.py:561:41

mov.u32         %r44, %tid.x;

shl.b32         %r45, %r44, 2;

and.b32         %r46, %r45, 508;

.loc    1 561 28                        // lecture_06.py:561:28

or.b32          %r47, %r42, %r46;

or.b32          %r48, %r47, 512;

.loc    1 564 21                        // lecture_06.py:564:21

setp.lt.s32     %p1, %r47, %r43;

setp.lt.s32     %p2, %r48, %r43;

.loc    1 567 24                        // lecture_06.py:567:24

mul.wide.s32    %rd7, %r47, 4;

add.s64         %rd1, %rd5, %rd7;

add.s64         %rd2, %rd1, 2048;

.loc    1 567 16                        // lecture_06.py:567:16

// begin inline asm

mov.u32 %r2, 0x0;

mov.u32 %r3, 0x0;

mov.u32 %r4, 0x0;

mov.u32 %r5, 0x0;

@%p1 ld.global.v4.b32 { %r2, %r3, %r4, %r5 }, [ %rd1 + 0 ];

// end inline asm

mov.b32         %f17, %r2;

mov.b32         %f18, %r3;

mov.b32         %f19, %r4;

mov.b32         %f20, %r5;

// begin inline asm

mov.u32 %r6, 0x0;

mov.u32 %r7, 0x0;

mov.u32 %r8, 0x0;

mov.u32 %r9, 0x0;

@%p2 ld.global.v4.b32 { %r6, %r7, %r8, %r9 }, [ %rd2 + 0 ];

// end inline asm

mov.b32         %f21, %r6;

mov.b32         %f22, %r7;

mov.b32         %f23, %r8;

mov.b32         %f24, %r9;

.loc    1 571 37                        // lecture_06.py:571:37

mul.f32         %f25, %f17, 0f3D372713;

mul.f32         %f26, %f18, 0f3D372713;

mul.f32         %f27, %f19, 0f3D372713;

mul.f32         %f28, %f20, 0f3D372713;

mul.f32         %f29, %f21, 0f3D372713;

mul.f32         %f30, %f22, 0f3D372713;

mul.f32         %f31, %f23, 0f3D372713;

mul.f32         %f32, %f24, 0f3D372713;

.loc    1 571 41                        // lecture_06.py:571:41

mul.f32         %f33, %f25, %f17;

mul.f32         %f34, %f26, %f18;

mul.f32         %f35, %f27, %f19;

mul.f32         %f36, %f28, %f20;

mul.f32         %f37, %f29, %f21;

mul.f32         %f38, %f30, %f22;

mul.f32         %f39, %f31, %f23;

mul.f32         %f40, %f32, %f24;

.loc    1 571 26                        // lecture_06.py:571:26

fma.rn.f32      %f41, %f33, %f17, %f17;

fma.rn.f32      %f42, %f34, %f18, %f18;

fma.rn.f32      %f43, %f35, %f19, %f19;

fma.rn.f32      %f44, %f36, %f20, %f20;

fma.rn.f32      %f45, %f37, %f21, %f21;

fma.rn.f32      %f46, %f38, %f22, %f22;

fma.rn.f32      %f47, %f39, %f23, %f23;

fma.rn.f32      %f48, %f40, %f24, %f24;

.loc    1 571 22                        // lecture_06.py:571:22

mul.f32         %f49, %f41, 0f3F4C422A;

mul.f32         %f50, %f42, 0f3F4C422A;

mul.f32         %f51, %f43, 0f3F4C422A;

mul.f32         %f52, %f44, 0f3F4C422A;

mul.f32         %f53, %f45, 0f3F4C422A;

mul.f32         %f54, %f46, 0f3F4C422A;

mul.f32         %f55, %f47, 0f3F4C422A;

mul.f32         %f56, %f48, 0f3F4C422A;

.loc    1 572 21                        // lecture_06.py:572:21

fma.rn.f32      %f57, %f41, 0f3F4C422A, %f49;

fma.rn.f32      %f58, %f42, 0f3F4C422A, %f50;

fma.rn.f32      %f59, %f43, 0f3F4C422A, %f51;

fma.rn.f32      %f60, %f44, 0f3F4C422A, %f52;

fma.rn.f32      %f61, %f45, 0f3F4C422A, %f53;

fma.rn.f32      %f62, %f46, 0f3F4C422A, %f54;

fma.rn.f32      %f63, %f47, 0f3F4C422A, %f55;

fma.rn.f32      %f64, %f48, 0f3F4C422A, %f56;

.loc    1 572 17                        // lecture_06.py:572:17

mul.f32         %f2, %f57, 0f3FB8AA3B;

// begin inline asm

ex2.approx.f32 %f1, %f2;

// end inline asm

mul.f32         %f4, %f58, 0f3FB8AA3B;

// begin inline asm

ex2.approx.f32 %f3, %f4;

// end inline asm

mul.f32         %f6, %f59, 0f3FB8AA3B;

// begin inline asm

ex2.approx.f32 %f5, %f6;

// end inline asm

mul.f32         %f8, %f60, 0f3FB8AA3B;

// begin inline asm

ex2.approx.f32 %f7, %f8;

// end inline asm

mul.f32         %f10, %f61, 0f3FB8AA3B;

// begin inline asm

ex2.approx.f32 %f9, %f10;

// end inline asm

mul.f32         %f12, %f62, 0f3FB8AA3B;

// begin inline asm

ex2.approx.f32 %f11, %f12;

// end inline asm

mul.f32         %f14, %f63, 0f3FB8AA3B;

// begin inline asm

ex2.approx.f32 %f13, %f14;

// end inline asm

mul.f32         %f16, %f64, 0f3FB8AA3B;

// begin inline asm

ex2.approx.f32 %f15, %f16;

// end inline asm

.loc    1 573 18                        // lecture_06.py:573:18

add.f32         %f65, %f1, 0fBF800000;
add.f32         %f66, %f3, 0fBF800000;
add.f32         %f67, %f5, 0fBF800000;
add.f32         %f68, %f7, 0fBF800000;
add.f32         %f69, %f9, 0fBF800000;
add.f32         %f70, %f11, 0fBF800000;
add.f32         %f71, %f13, 0fBF800000;
add.f32         %f72, %f15, 0fBF800000;
.loc    1 573 30                        // lecture_06.py:573:30
add.f32         %f73, %f1, 0f3F800000;
add.f32         %f74, %f3, 0f3F800000;
add.f32         %f75, %f5, 0f3F800000;
add.f32         %f76, %f7, 0f3F800000;
add.f32         %f77, %f9, 0f3F800000;
add.f32         %f78, %f11, 0f3F800000;
add.f32         %f79, %f13, 0f3F800000;
add.f32         %f80, %f15, 0f3F800000;
.loc    1 573 24                        // lecture_06.py:573:24
mov.b32         %r11, %f65;
mov.b32         %r12, %f73;
// begin inline asm
div.full.f32 %r10, %r11, %r12;
// end inline asm
mov.b32         %f81, %r10;
mov.b32         %r14, %f66;
mov.b32         %r15, %f74;
// begin inline asm
div.full.f32 %r13, %r14, %r15;
// end inline asm
mov.b32         %f82, %r13;
mov.b32         %r17, %f67;
mov.b32         %r18, %f75;
// begin inline asm
div.full.f32 %r16, %r17, %r18;
// end inline asm
mov.b32         %f83, %r16;
mov.b32         %r20, %f68;
mov.b32         %r21, %f76;
// begin inline asm
div.full.f32 %r19, %r20, %r21;
// end inline asm
mov.b32         %f84, %r19;
mov.b32         %r23, %f69;
mov.b32         %r24, %f77;
// begin inline asm
div.full.f32 %r22, %r23, %r24;
// end inline asm
mov.b32         %f85, %r22;
mov.b32         %r26, %f70;
mov.b32         %r27, %f78;
// begin inline asm
div.full.f32 %r25, %r26, %r27;
// end inline asm
mov.b32         %f86, %r25;
mov.b32         %r29, %f71;
mov.b32         %r30, %f79;
// begin inline asm
div.full.f32 %r28, %r29, %r30;
// end inline asm
mov.b32         %f87, %r28;
mov.b32         %r32, %f72;
mov.b32         %r33, %f80;
// begin inline asm
div.full.f32 %r31, %r32, %r33;
// end inline asm
mov.b32         %f88, %r31;
.loc    1 574 14                        // lecture_06.py:574:14
mul.f32         %f89, %f17, 0f3F000000;
mul.f32         %f90, %f18, 0f3F000000;
mul.f32         %f91, %f19, 0f3F000000;
mul.f32         %f92, %f20, 0f3F000000;
mul.f32         %f93, %f21, 0f3F000000;
mul.f32         %f94, %f22, 0f3F000000;
mul.f32         %f95, %f23, 0f3F000000;
mul.f32         %f96, %f24, 0f3F000000;
.loc    1 574 23                        // lecture_06.py:574:23
add.f32         %f97, %f81, 0f3F800000;
add.f32         %f98, %f82, 0f3F800000;
add.f32         %f99, %f83, 0f3F800000;
add.f32         %f100, %f84, 0f3F800000;
add.f32         %f101, %f85, 0f3F800000;
add.f32         %f102, %f86, 0f3F800000;
add.f32         %f103, %f87, 0f3F800000;
add.f32         %f104, %f88, 0f3F800000;
.loc    1 574 19                        // lecture_06.py:574:19
mul.f32         %f105, %f89, %f97;
mul.f32         %f106, %f90, %f98;
mul.f32         %f107, %f91, %f99;
mul.f32         %f108, %f92, %f100;
mul.f32         %f109, %f93, %f101;
mul.f32         %f110, %f94, %f102;
mul.f32         %f111, %f95, %f103;
mul.f32         %f112, %f96, %f104;
.loc    1 577 21                        // lecture_06.py:577:21
add.s64         %rd3, %rd6, %rd7;
add.s64         %rd4, %rd3, 2048;
.loc    1 577 30                        // lecture_06.py:577:30
mov.b32         %r34, %f105;
mov.b32         %r35, %f106;
mov.b32         %r36, %f107;
mov.b32         %r37, %f108;
// begin inline asm
@%p1 st.global.v4.b32 [ %rd3 + 0 ], { %r34, %r35, %r36, %r37 };
// end inline asm
mov.b32         %r38, %f109;
mov.b32         %r39, %f110;
mov.b32         %r40, %f111;
mov.b32         %r41, %f112;
// begin inline asm
@%p2 st.global.v4.b32 [ %rd4 + 0 ], { %r38, %r39, %r40, %r41 };
// end inline asm
.loc    1 577 4                         // lecture_06.py:577:4
ret;
$L__tmp1:$L__func_end0:
                                        // -- End function
}
.file   1 "/home/c-thashim/2025/spring2025-lectures/lecture_06.py"
.section        .debug_abbrev
{
.b8 1                                   // Abbreviation Code.b8 17                                  // DW_TAG_compile_unit.b8 0                                   //
}
.section        .debug_info
{
.b32 76                                 // Length of Unit.b8 2                                  // DWARF version number.b8 0.b32 .debug_abbrev
}
.section        .debug_macinfo  {       }
```

Observations:

- ld.global.* and st.global.* reads and writes from global memory
- %ctaid.x is block index, %tid.x is thread index
- %f* are floating point registers, %r* are integer registers
- One thread processes 8 elements at the same time (thread coarsening)

```python
def print_ptx(name: str, kernel):
    if os.environ.get("TRITON_INTERPRET") == "1":
        text("PTX is not generated when in interpret mode.")
        return

    """Print out the PTX code generated by Triton for the given `kernel`."""
    ptx_path = f"var/{name}-ptx.txt"
```

Let's go poke around at the PTX code.
https://github.com/stanford-cs336/spring2025-lectures/blob/main/var/triton_softmax-ptx.txt

```python
    with open(ptx_path, "w") as f:
        return list(kernel.cache[0].values())[0].asm["ptx"]




def pytorch_compilation():
```

So far, we have seen three ways to write GeLU:

- Use the default PyTorch function
- Write it in Python manual_gelu
- Write it in CUDA create_cuda_gelu
- Write it in Triton triton_gelu

- Write it in Python and compile it into Triton

```python
compiled_gelu = torch.compile(manual_gelu)
```

Check correctness of our implementation.

```python
check_equal(compiled_gelu, manual_gelu)

if not torch.cuda.is_available():
    return
```

Let's benchmark and profile it!

```python
manual_time = benchmark("manual_gelu", run_operation1(dim=16384, operation=manual_gelu)) # @inspect manual_time
pytorch_time = benchmark("pytorch_gelu", run_operation1(dim=16384, operation=pytorch_gelu)) # @inspect pytorch_time
cuda_time = benchmark("cuda_gelu", run_operation1(dim=16384, operation=create_cuda_gelu())) # @inspect cuda_time
triton_time = benchmark("triton_gelu", run_operation1(dim=16384, operation=triton_gelu)) # @inspect triton_time
compiled_time = benchmark("compiled_gelu", run_operation1(dim=16384, operation=compiled_gelu)) # @inspect compiled_time
```

Let's look under the hood

```python
compiled_gelu_profile = profile("compiled_gelu", run_operation1(dim=16384, operation=compiled_gelu))
```

compiled_gelu

```
                        Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Self CUDA %    CUDA total  CUDA time avg    # o
------------------------------------------------------------------------------------------
      Torch-Compiled Region: 0/1        77.92%       1.934ms        91.35%       2.268ms       2.268ms       0.000us         0.00%     707.261us     707.261us
triton_poi_fused_add_mul_tanh_0         1.84%      45.795us        13.43%     333.455us     333.455us     707.261us       100.00%     707.261us     707.261us
triton_poi_fused_add_mul_tanh_0         0.00%       0.000us         0.00%       0.000us       0.000us     707.261us       100.00%     707.261us     707.261us
         TorchDynamo Cache Lookup         0.62%      15.371us         0.62%      15.371us      15.371us       0.000us         0.00%       0.000us       0.000us
                   cuLaunchKernel        11.59%     287.660us        11.59%     287.660us     287.660us       0.000us         0.00%       0.000us       0.000us
            cudaDeviceSynchronize         8.03%     199.299us         8.03%     199.299us      99.649us       0.000us         0.00%       0.000us       0.000us
------------------------------------------------------------------------------------------
 Self CPU time total: 2.482msSelf CUDA time total: 707.261us
```

```python
def triton_softmax_main():
```

So far, we've looked at elementwise operations in Triton (e.g., GeLU).
Now let us look at operations that aggregate over multiple values.

We will roughly follow the Triton fused softmax tutorial: https://triton-lang.org/main/getting-started/tutorials/02-fused-softmax.html

Recall the softmax operation is used in attention and generating probabilities.
Normalize each row of a matrix:

```
[A1 A2 A3]   =>   [A1/A A2/A A3/A]
[B1 B2 B3]   =>   [B1/B B2/B B3/B]
```

Let's first start with the naive implementation and keep track of reads/writes.

```python
x = torch.tensor([
    [5., 5, 5],
    [0, 0, 100],
], device=get_device())
y1 = manual_softmax(x) # @inspect y1

if not torch.cuda.is_available():
    return
```

Now let us write the Triton kernel.

```python
y2 = triton_softmax(x)
assert torch.allclose(y1, y2)
```

Check our implementations are correct.

```python
check_equal2(pytorch_softmax, manual_softmax)
check_equal2(pytorch_softmax, triton_softmax)

compiled_softmax = torch.compile(manual_softmax)
```

Now let's benchmark everything.

```python
manual_time = benchmark("manual_softmax", run_operation1(dim=16384, operation=manual_softmax)) # @inspect manual_time
compiled_time = benchmark("compiled_softmax", run_operation1(dim=16384, operation=compiled_softmax)) # @inspect compiled_time
pytorch_time = benchmark("pytorch_softmax", run_operation1(dim=16384, operation=pytorch_softmax)) # @inspect pytorch_time
triton_time = benchmark("triton_softmax", run_operation1(dim=16384, operation=triton_softmax)) # @inspect triton_time
```

Look under the hood using the profiler.

```python
manual_softmax_profile = profile("manual_softmax", run_operation1(dim=16384, operation=manual_softmax))
```

manual_softmax

```
                                                                                   Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se
------------------------------------------------------------------------------------------
                                                                              aten::div         0.28%       8.466us         0.43%      13.241us      13.241us     921.170us
 void at::native::elementwise_kernel<128, 2, at::native::gpu_kernel_impl_nocas...         0.00%       0.000us         0.00%       0.000us       0.000us     921.170us
                                                                              aten::sub         0.44%      13.364us         0.70%      21.601us      21.601us     892.243us
 void at::native::elementwise_kernel<128, 2, at::native::gpu_kernel_impl_nocas...         0.00%       0.000us         0.00%       0.000us       0.000us     892.243us
                                                                              aten::exp         0.25%       7.610us         0.41%      12.595us      12.595us     703.925us
 void at::native::vectorized_elementwise_kernel<4, at::native::exp_kernel_cuda...         0.00%       0.000us         0.00%       0.000us       0.000us     703.925us
                                                                              aten::max        10.71%     328.962us        19.95%     612.839us     612.839us     393.433us
 void at::native::reduce_kernel<512, 1, at::native::ReduceOp<float, at::native...         0.00%       0.000us         0.00%       0.000us       0.000us     393.433us
                                                                              aten::sum         0.38%      11.798us         0.67%      20.683us      20.683us     346.779us
 void at::native::reduce_kernel<512, 1, at::native::ReduceOp<float, at::native...         0.00%       0.000us         0.00%       0.000us       0.000us     346.779us
------------------------------------------------------------------------------------------
 Self CPU time total: 3.071msSelf CUDA time total: 3.258ms
```

```python
compiled_softmax_profile = profile("compiled_softmax", run_operation1(dim=16384, operation=compiled_softmax))
```

compiled_softmax

```
                           Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Self CUDA %    CUDA total  CUDA time avg
------------------------------------------------------------------------------------------
         Torch-Compiled Region: 1/0        56.77%     769.985us        78.99%       1.071ms       1.071ms       0.000us         0.00%     730.770us     730.770us
triton_red_fused_div_exp_max_sub_sum_0         3.20%      43.382us        22.22%     301.366us     301.366us     730.770us       100.00%     730.770us     730.770us
triton_red_fused_div_exp_max_sub_sum_0         0.00%       0.000us         0.00%       0.000us       0.000us     730.770us       100.00%     730.770us     730.770us
               TorchDynamo Cache Lookup         0.53%       7.239us         0.53%       7.239us       7.239us       0.000us         0.00%       0.000us       0.000us
                         cuLaunchKernel        19.02%     257.984us        19.02%     257.984us     257.984us       0.000us         0.00%       0.000us       0.000us
                  cudaDeviceSynchronize        20.48%     277.800us        20.48%     277.800us     138.900us       0.000us         0.00%       0.000us       0.000us
------------------------------------------------------------------------------------------
 Self CPU time total: 1.356msSelf CUDA time total: 730.770us
```

```python
pytorch_softmax_profile = profile("pytorch_softmax", run_operation1(dim=16384, operation=pytorch_softmax))
```

pytorch_softmax

```
                                                                                   Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Se
------------------------------------------------------------------------------------------
                                                                          aten::softmax         0.47%       5.061us        28.87%     312.564us     312.564us       0.000us
                                                                         aten::_softmax        13.44%     145.534us        28.40%     307.503us     307.503us       1.137ms
 void at::native::(anonymous namespace)::cunn_SoftMaxForward<4, float, float, ...         0.00%       0.000us         0.00%       0.000us       0.000us       1.137ms
                                                                       cudaLaunchKernel        14.96%     161.969us        14.96%     161.969us     161.969us       0.000us
                                                                  cudaDeviceSynchronize        71.13%     770.228us        71.13%     770.228us     385.114us       0.000us
------------------------------------------------------------------------------------------
 Self CPU time total: 1.083msSelf CUDA time total: 1.137ms
```

```python
triton_softmax_profile = profile("triton_softmax", run_operation1(dim=16384, operation=triton_softmax))
```

triton_softmax

```
                  Name    Self CPU %      Self CPU   CPU total %     CPU total  CPU time avg     Self CUDA   Self CUDA %    CUDA total  CUDA time avg    # of Calls
------------------------------------------------------------------------------------------
triton_softmax_kernel         0.00%       0.000us         0.00%       0.000us       0.000us     705.462us       100.00%     705.462us     705.462us             1
     aten::empty_like         0.23%       4.238us        77.57%       1.409ms       1.409ms       0.000us         0.00%       0.000us       0.000us             1
  aten::empty_strided        77.34%       1.405ms        77.34%       1.405ms       1.405ms       0.000us         0.00%       0.000us       0.000us             1
       cuLaunchKernel         8.02%     145.738us         8.02%     145.738us     145.738us       0.000us         0.00%       0.000us       0.000us             1
cudaDeviceSynchronize        14.41%     261.640us        14.41%     261.640us     130.820us       0.000us         0.00%       0.000us       0.000us             2
------------------------------------------------------------------------------------------
 Self CPU time total: 1.816msSelf CUDA time total: 705.462us
```

Let's end by looking at the PTX code.

```python
ptx = print_ptx("triton_softmax", triton_softmax_kernel)
```

```
//// Generated by LLVM NVPTX Back-End//.version 8.4.target sm_90a.address_size 64
// .globl       triton_softmax_kernel   // -- Begin function triton_softmax_kernel
                                        // @triton_softmax_kernel
.visible .entry triton_softmax_kernel(
.param .u64 .ptr .global .align 1 triton_softmax_kernel_param_0,
.param .u64 .ptr .global .align 1 triton_softmax_kernel_param_1,
.param .u32 triton_softmax_kernel_param_2,
.param .u32 triton_softmax_kernel_param_3,
.param .u32 triton_softmax_kernel_param_4
).reqntid 128, 1, 1{
.reg .pred      %p<5>;
.reg .b32       %r<22>;
.reg .f32       %f<13>;
.reg .b64       %rd<10>;
.loc    1 741 0                         // lecture_06.py:741:0
$L__func_begin0:
.loc    1 741 0                         // lecture_06.py:741:0
// %bb.0:
ld.param.u64    %rd3, [triton_softmax_kernel_param_0];
ld.param.u64    %rd4, [triton_softmax_kernel_param_1];
$L__tmp0:
.loc    1 745 28                        // lecture_06.py:745:28
// begin inline asm
mov.u32 %r1, %ctaid.x;
// end inline asm
ld.param.u32    %r8, [triton_softmax_kernel_param_2];
.loc    1 746 31                        // lecture_06.py:746:31
mov.u32         %r9, %tid.x;
and.b32         %r10, %r9, 3;
ld.param.u32    %r11, [triton_softmax_kernel_param_3];
.loc    1 749 36                        // lecture_06.py:749:36
mul.lo.s32      %r12, %r1, %r8;
ld.param.u32    %r13, [triton_softmax_kernel_param_4];
.loc    1 749 26                        // lecture_06.py:749:26
mul.wide.s32    %rd5, %r12, 4;
add.s64         %rd6, %rd3, %rd5;
.loc    1 750 27                        // lecture_06.py:750:27
mul.wide.u32    %rd7, %r10, 4;
add.s64         %rd1, %rd6, %rd7;
.loc    1 751 47                        // lecture_06.py:751:47
setp.lt.s32     %p1, %r10, %r13;
mov.b32         %r3, -8388608;
.loc    1 751 20                        // lecture_06.py:751:20
// begin inline asm
mov.u32 %r2, 0x0;
@%p1 ld.global.b32 { %r2 }, [ %rd1 + 0 ];
@!%p1 mov.u32 %r2, %r3;
// end inline asm
mov.b32         %f3, %r2;
$L__tmp1:
.loc    2 184 40                        // standard.py:184:40
shfl.sync.bfly.b32      %r14, %r2, 2, 31, -1;
mov.b32         %f4, %r14;
.loc    2 163 27                        // standard.py:163:27
max.f32         %f5, %f3, %f4;
.loc    2 184 40                        // standard.py:184:40
mov.b32         %r15, %f5;
shfl.sync.bfly.b32      %r16, %r15, 1, 31, -1;
mov.b32         %f6, %r16;
.loc    2 163 27                        // standard.py:163:27
max.f32         %f7, %f5, %f6;
$L__tmp2:
.loc    1 754 20                        // lecture_06.py:754:20
sub.f32         %f8, %f3, %f7;
.loc    1 755 23                        // lecture_06.py:755:23
mul.f32         %f2, %f8, 0f3FB8AA3B;
// begin inline asm
ex2.approx.f32 %f1, %f2;
// end inline asm
$L__tmp3:
.loc    2 267 36                        // standard.py:267:36
mov.b32         %r5, %f1;
shfl.sync.bfly.b32      %r17, %r5, 2, 31, -1;
mov.b32         %f9, %r17;
.loc    2 256 15                        // standard.py:256:15
add.f32         %f10, %f1, %f9;
.loc    2 267 36                        // standard.py:267:36
mov.b32         %r18, %f10;
shfl.sync.bfly.b32      %r19, %r18, 1, 31, -1;
mov.b32         %f11, %r19;
.loc    2 256 15                        // standard.py:256:15
add.f32         %f12, %f10, %f11;
$L__tmp4:
.loc    1 757 24                        // lecture_06.py:757:24
mov.b32         %r6, %f12;
// begin inline asm
div.full.f32 %r7, %r5, %r6;
// end inline asm
.loc    1 760 36                        // lecture_06.py:760:36
mul.lo.s32      %r20, %r1, %r11;
.loc    1 760 26                        // lecture_06.py:760:26
mul.wide.s32    %rd8, %r20, 4;
add.s64         %rd9, %rd4, %rd8;
.loc    1 761 27                        // lecture_06.py:761:27
add.s64         %rd2, %rd9, %rd7;
.loc    1 762 21                        // lecture_06.py:762:21
and.b32         %r21, %r9, 124;
setp.eq.s32     %p4, %r21, 0;
and.pred        %p3, %p4, %p1;
// begin inline asm
@%p3 st.global.b32 [ %rd2 + 0 ], { %r7 };
// end inline asm
.loc    1 762 4                         // lecture_06.py:762:4
ret;
$L__tmp5:$L__func_end0:
                                        // -- End function
}
.file   1 "/home/c-thashim/2025/spring2025-lectures/lecture_06.py"
.file   2 "/home/c-thashim/2025/spring2025-lectures/.venv/lib/python3.10/site-packages/triton/language/standard.py"
.section        .debug_abbrev
{
.b8 1                                   // Abbreviation Code.b8 17                                  // DW_TAG_compile_unit.b8 1                                   //
}
.section        .debug_info
{
.b32 173                                // Length of Unit.b8 2                                  // DWARF version number.b8 0.b32 .debug_abbrev
}
.section        .debug_macinfo  {       }
```

```python
def manual_softmax(x: torch.Tensor):
    # M: number of rows, N: number of columns
    M, N = x.shape

    # Compute the max of each row (MN reads, M writes)
    x_max = x.max(dim=1)[0]

    # Subtract off the max (MN + M reads, MN writes)
    x = x - x_max[:, None]

    # Exponentiate (MN reads, MN writes)
    numerator = torch.exp(x)

    # Compute normalization constant (MN reads, M writes)
    denominator = numerator.sum(dim=1)

    # Normalize (MN reads, MN writes)
    y = numerator / denominator[:, None]

    # Total: 5MN + M reads, 3MN + 2M writes
    # In principle, should have MN reads, MN writes (speedup of 4x!)
    return y


def triton_softmax(x: torch.Tensor):
    # Allocate output tensor
    y = torch.empty_like(x)

    # Determine grid
    M, N = x.shape                          # Number of rows x number of columns
    block_size = triton.next_power_of_2(N)  # Each block contains all the columns
    num_blocks = M                          # Each block is a row

    # Launch kernel
    triton_softmax_kernel[(M,)](
        x_ptr=x, y_ptr=y,
        x_row_stride=x.stride(0), y_row_stride=y.stride(0),
        num_cols=N, BLOCK_SIZE=block_size
    )

    return y


@triton.jit
def triton_softmax_kernel(x_ptr, y_ptr, x_row_stride, y_row_stride, num_cols, BLOCK_SIZE: tl.constexpr):
    assert num_cols <= BLOCK_SIZE

    # Process each row independently
    row_idx = tl.program_id(0)
    col_offsets = tl.arange(0, BLOCK_SIZE)

    # Read from global memory
    x_start_ptr = x_ptr + row_idx * x_row_stride
    x_start_ptr = x_ptr + row_idx * x_row_stride
    x_ptrs = x_start_ptr + col_offsets
    x_row = tl.load(x_ptrs, mask=col_offsets < num_cols, other=float("-inf"))

    # Compute
    x_row = x_row - tl.max(x_row, axis=0)
    numerator = tl.exp(x_row)
    denominator = tl.sum(numerator, axis=0)
    y_row = numerator / denominator

    # Write back to global memory
    y_start_ptr = y_ptr + row_idx * y_row_stride
    y_ptrs = y_start_ptr + col_offsets
    tl.store(y_ptrs, y_row, mask=col_offsets < num_cols)


def triton_matmul_main():
    text("Matrix multipliction is perhaps the most optimized algorithm ever.")

    text("If you write matrix multiplication in CUDA, there's all sorts of crazy things you have to do.")
    link("https://github.com/openai/blocksparse/blob/master/src/matmul_op_gpu.cu")

    text("It's much easier in Triton.")
    link("https://triton-lang.org/main/getting-started/tutorials/03-matrix-multiplication.html")

    text("       k                 j                    ", verbatim=True)
    text("  [ A1 A2 A3 ]       [ B1 B2 B3 ]   [ C1 C2 C3 ]", verbatim=True)
    text("i [ A4 A5 A6 ]  *  k [ B4 B5 B6 ] = [ C4 C5 C6 ]", verbatim=True)
    text("  [ A7 A8 A9 ]       [ B7 B8 B9 ]   [ C7 C8 C9 ]", verbatim=True)

    text("Naively: need MKN reads, MN writes")

    text("Computing C4 and C5 both need A4, A5, A6.")
    text("Can we read A4, A5, A6 from DRAM once to compute both?")
    text("Answer: yes, using shared memory!")

    text("## Tiling (leveraging shared memory)")

    text("Recall that shared memory is:")
    text("- fast (10x faster) and small(~100KB)")
    text("- shared between all the threads in a block.")
    image("https://miro.medium.com/v2/resize:fit:2000/format:webp/1*6xoBKi5kL2dZpivFe1-zgw.jpeg")

    text("Trivial: for small matrices, load all of A and B into shared memory, then could compute C.")
    text("Now we get MK + KN reads, MN writes")

    text("But what if we have big matrices...")

    image("https://www.researchgate.net/profile/Axel-Huebl/publication/320499173/figure/fig1/AS:614298980196359@1523471698396/Performance-critical-A-B-part-of-the-GEMM-using-a-tiling-strategy-A-thread-iterates.png", width=0.5)
    text("Key idea: divide the matrix into blocks.")
    text("For each block of A and block of B:")
    text("- load into shared memory,")
    text("- do mini-matrix multiplication,")
    text("- write the partial sum.")

    text("Animation of tiled matrix multiplication "), link("https://youtu.be/aMvCEEBIBto")

    text("## Leveraging L2 cache")

    text("Two ways of computing 9 elements of a matrix:")
    image("https://triton-lang.org/main/_images/grouped_vs_row_major_ordering.png", width=0.5)
    text("1. Loads 9 + 81 = 90 blocks")
    text("1. Loads 27 + 27 = 54 blocks")

    text("Process the blocks in an order that minimizes the reads.")

    text("Why write your own kernel for matrix multiplication (e.g., A @ B)?")
    text("Answer: fusion with another operation (e.g., gelu(A @ B))")

    if not torch.cuda.is_available():
        return
    text("Let's try it!")
    benchmark("pytorch_matmul", run_operation2(dim=16384, operation=torch.matmul))
    benchmark("triton_matmul", run_operation2(dim=16384, operation=triton_matmul))

    # Not working for some reason
    #print_ptx("triton_matmul", triton_matmul_kernel)


def further_reading():
    Horace He's blog post  [Article]

    CUDA MODE Lecture 1: how to profile CUDA kernels in PyTorch  [Video]
    CUDA MODE Lecture 2: Chapters 1-3 of PPMP book  [Video]
    CUDA MODE Lecture 3: Getting started with CUDA for Python Programmers  [Video]
    CUDA MODE Lecture 4: Compute and memory basics  [Video]
    CUDA MODE Lecture 8: CUDA performance checklist  [Video]

    HetSys Course: Lecture 1: Programming heterogenous computing systems with GPUs  [Video]
    HetSys Course: Lecture 2: SIMD processing and GPUs  [Video]
    HetSys Course: Lecture 3: GPU Software Hierarchy  [Video]
    HetSys Course: Lecture 4: GPU Memory Hierarchy  [Video]
    HetSys Course: Lecture 5: GPU performance considerations  [Video]

    [A100 GPU with NVIDIA Ampere Architecture]
    [NVIDIA Deep Learning Performance Guide]
    [GPU Puzzles]
    [Triton Paper]
    [PyTorch 2.0 Acceleration]

############################################################

def print_gpu_specs():
    num_devices = torch.cuda.device_count()  # @inspect num_devices
    8 devices
    for i in range(num_devices):
        properties = torch.cuda.get_device_properties(i)  # @inspect properties
        7: _CudaDeviceProperties(name='NVIDIA H100 80GB HBM3', major=9, minor=0, total_memory=81090MB,
        multi_processor_count=132, uuid=62f395b0-f63d-2a9d-d202-53f798ada4f4, L2_cache_size=50MB)


def pytorch_softmax(x: torch.Tensor):
    return torch.nn.functional.softmax(x, dim=-1)


def pytorch_gelu(x: torch.Tensor):
    # Use the tanh approximation to match our implementation
    return torch.nn.functional.gelu(x, approximate="tanh")


def manual_gelu(x: torch.Tensor):
    return 0.5 * x * (1 + torch.tanh(0.79788456 * (x + 0.044715 * x * x * x)))




if __name__ == "__main__":
    main()
```